

Hypervector Frames for Category-Guided Causal Coding: Merging HDC Capacity Theory with Universals-Guided Transformer Training, with a Staged Small-Scale Measurement Programme

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Abstract

Two recently developed frameworks approach the internal organization of transformer networks from opposite directions. Category-Guided Causal-Coding Transformers (CGCCT) treat a trained network as a quantale-enriched category of modular latent states and context updates, connect it to language through learned cognition-to-frame and frame-to-language maps, and use typological universals—implicational hierarchies, closure laws, naturality squares—as training signals that shape latent causal modules. The theory of resonator-factored hierarchical hypervector codes (HDC) supplies a concrete algebra of role-bound factor representations together with a quantitative capacity theory: coherence-aware cleanup bounds, basin-conditional resonator decoding, admissibility diagnostics, and losses that push a host learner toward low-load, high-margin, role-stable factorizations. This note argues that the two frameworks compose: hypervector frame codes give CGCCT’s frame category a computable object language in which naturality, closure, mediator structure, do-footprints, and double-counting become differentiable algebraic quantities, while CGCCT gives the HDC admissibility losses a principled organizing grammar and a source of linguistically grounded supervision. The composition also transfers guarantees: CGCCT’s probe-calibration assumption is discharged by the HDC coherence-aware capacity law, and we conjecture that implicational hierarchies sit precisely in the law’s high-coherence (near-duplicate) regime, making the corrected $k^{3/2}$ – k^2 dimension scaling operative rather than optional. We lay out which quantities of the merged framework are measurable at small scale—synthetic grammars, GPT-2 small, Pythia-160M checkpoints, small multilingual models—and which are not, and we provide, as an appendix, a detailed staged testing programme at GPT-2-small scale written for autonomous coding agents, with validation gates, known-answer checks (including rediscovery of the indirect-object-identification circuit), frozen design choices, compute budgets, and stop conditions.

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Citation caution External citations in this note were recalled from model memory during drafting and must be verified against the literature before being relied upon or propagated.

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1 Introduction

CGCCT and the HDC capacity framework were developed for different proximate purposes. CGCCT asks how the latent organization of a transformer can be *shaped*: it equips the network’s modular latent states and context updates with a quantale-enriched categorical structure CT_Q , posits learned maps

$$\text{CT}_Q \xrightarrow{\hat{K}} \text{Frame}_Q \xrightarrow{\hat{E}} \text{Lang}_Q, \quad \hat{\Lambda} = \hat{E} \circ \hat{K}, \quad (1)$$

and turns typological universals—implicational hierarchies, closure under composition, cross-context naturality—into training losses on modules, mediators, and update commutators. The HDC framework asks how factorized latent structure can be *stored, queried, and protected*: it provides a fixed vector-symbolic algebra of role-bound factor codes, a coherence-aware capacity theory saying when cleanup and resonator decoding are reliable, and a family of admissibility diagnostics and losses—effective load, cleanup margin, confusable-set size, dictionary coherence, role stability, product

recomposability, residual synergy, intervention locality—that press an upstream learner toward representations the algebra can serve.

The claim of this note is that each framework supplies exactly what the other leaves underdetermined.

On the CGCCT side, three load-bearing components are specified abstractly but not concretely. The frame category Frame_Q needs an object language: something that is structured enough to support role maps, refinements, and restrictions as morphisms, yet continuous enough to train against and to measure with. The naturality loss needs a metric d on frame and linguistic objects and an executable implementation of the linguistic-side transforms V_c . And the closure-preservation guarantee leans on δ -calibrated probes whose calibration is, in the original presentation, an empirical assumption. Hypervector frame codes answer all three: typed role-filler hypervectors are the object language; algebraic rebinding, substitution, and masking implement the morphisms and the V_c ; similarity is the metric; and probe calibration is not assumed but *bounded* by the coherence-aware capacity law, with an explicit dimension prescription.

On the HDC side, the admissibility losses are individually motivated but collectively ad hoc: nothing in the capacity theory says which role inventories, which factor types, or which invariances a language system should be pressed toward. CGCCT supplies that grammar. Universals fix which probes must exist and which implications must close; alternation phenomena fix a family of naturality squares with *known* algebraic right-hand sides; the mediator and double-counting rules fix how cross-module structure must be routed and audited. The HDC role-consistency loss turns out to be a single naturality square; CGCCT is the functorial family it belongs to.

The two frameworks also share their principal risk, and it is worth stating at the outset rather than discovering later: both presuppose that a low-defect frame-level factorization of transformer language states exists for the query families of interest. Neither the categorical machinery nor the hypervector algebra manufactures such a factorization; both can only detect, measure, and exploit it. What the merger adds is that the risk becomes a set of numbers with predicted parameter dependence—naturality defect, probe margin, closure-violation rate, factorization defect—rather than a philosophical worry. The measurement programme of Section 7 and Appendix A is organized around measuring those numbers cheaply before committing to expensive training.

2 The two frameworks in brief

2.1 CGCCT

CGCCT models a trained transformer as a quantale-enriched category CT_Q : objects are (typically modular) latent states $Z \cong Z_1 \otimes \cdots \otimes Z_M$ over a module inventory (heads, channels, adapters, columns), and morphisms are context updates U_c with sparse supports S_c and causal gates estimated from do-influence. A frame category Frame_Q holds semantic frames—typed role structures with morphisms for role mappings, refinements, and context restrictions—and a linguistic category Lang_Q holds externalized forms. The learned functors of Eq. (1) are trained with: a *naturality* loss asking $\hat{\Lambda}(U_c z) \approx V_c \hat{\Lambda}(z)$ for corresponding linguistic transforms V_c ; *closure* losses enforcing implicational hierarchies (if the language/state realizes feature a it must realize b); *mediator* losses forbidding dense direct coupling between modules that should communicate through a sparse mediator; *commutator* losses weighted by frame and linguistic overlap between contexts; *double-counting* penalties on pairs of modules with redundant causal footprints; and *stability scheduling* that maps the stability of a universal over training to the plasticity of the modules realizing it. Universals are subobjects $U \hookrightarrow L$ of linguistic objects, and a cognitive explanation of a universal is a pullback: the preimage $\hat{\Lambda}^{-1}(U)$ should align with a stable module structure rather than a scattered accidental

region. Module and mediator selection is governed by a multi-objective *weakness* order valued in a product quantale over fit, retention, commutator defect, leakage, double-counting, and cost.

2.2 The HDC capacity framework

The HDC framework fixes a vector-symbolic algebra (binding $\otimes$, bundling $\oplus$, permutation, cleanup against explicit dictionaries) and proves query-limited capacity results for hierarchical role-bound codes. The components used in this note are: the coherence-aware cleanup law, by which a single cleanup against a dictionary of size M with coherence ν under crosstalk load k succeeds except with probability $\sim Me^{-cD(1-\nu)/k}$, so that

$$D \gtrsim \frac{k}{c(1-\nu)} \log \frac{M}{\varepsilon}; \quad (2)$$

the near-duplicate analysis, by which structurally adjacent dictionary entries (differing in one of k role-bound constituents) force $D \gtrsim k^{3/2} \log M$ for sign-quantized codes and $D \gtrsim k^2 \log M$ for linear or phasor codes; basin-conditional resonator decoding of unknown factor products, with warm starts as the practical basin-entry mechanism; the admissibility diagnostics $(\delta_Q, k^{\text{eff}}, M^{\text{conf}}, \nu, \Delta, RKI, \eta_{\text{bind}})$ and the capacity proxy

$$\mathcal{C}_{\text{HDC}} = \max_t \frac{k_t^{\text{eff}}}{1-\nu_t} [\log M_t^{\text{conf}} + \log(q_{\text{max}} \ell_{\text{max}}/\varepsilon)]; \quad (3)$$

the Hierarchical Modular Hypervector (HMH) representation of typed role-filler structures with ordered argument roles, in which directed relations are kept oriented by applying a permutation (or dedicated source/target roles) to exactly one argument; and the factorization-defect discipline, by which the gap between full-state readouts and factor-graph readouts is measured per query family and treated as an irreducible floor that no coding dimension can remove.

3 Hypervector frames as the object language of Frame_Q

3.1 Frame codes

Let a frame instance consist of a frame type σ , a set of typed roles $\{\rho_1, \dots, \rho_m\}$ drawn from a fixed inventory, fillers $f_1, \dots, f_m$, and typed directed relations among role positions. Its hypervector code is the HMH form

$$F = \text{Normalize} \left(r_{\text{type}} \otimes h_{\sigma} \oplus \bigoplus_{i=1}^m r_{\rho_i} \otimes h_{f_i} \oplus \bigoplus_{(i,j,\lambda)} r_{\lambda} \otimes (r_{\rho_i} \otimes h_{f_i}) \otimes \pi(r_{\rho_j} \otimes h_{f_j}) \right), \quad (4)$$

with π a fixed permutation marking the target argument. Three properties make this the right object language for Frame_Q .

First, *morphisms are executable*. A role mapping $\rho_i \mapsto \rho'_i$ is the rebinding $F \mapsto F$ with $r_{\rho_i} \otimes h_{f_i}$ replaced by $r_{\rho'_i} \otimes h_{f_i}$ (implemented by subtract-and-add in real-sum systems, or by rebuilding the bundle from the frame record in sign or phasor systems). A refinement adds bound terms; a context restriction is a role mask followed by renormalization; a filler substitution is a slot swap. Each categorical arrow of Frame_Q becomes a concrete, differentiable-or- piecewise-constant operation on codes, so functoriality claims can be *tested*, not merely asserted.

Second, *enrichment comes for free*. Frame_Q is quantale-enriched; the hom-values need a truth-degree structure. Normalized similarity $\langle F, F' \rangle / D$, or margin-calibrated cleanup confidence, supplies hom-values in $[0, 1]$ with the right compositional behavior (composition can only lose similarity, matching the quantale multiplication’s monotonicity), and the capacity theory says when those values are trustworthy.

Third, *orientation is preserved, and its absence is detectable*. Under a commutative binding, the symmetric encoding of a two-argument frame is invariant under exchanging agent and patient: “dog bites man” and “man bites dog” would receive the same code, exactly as an unordered episodic encoding conflates $\text{causes}(a, b)$ with $\text{causes}(b, a)$. The permutation in Eq. (4) is therefore not a convention but a correctness condition, and it comes with an experimental signature: with the permutation ablated, the codes of argument-exchanged frames have similarity exactly 1; with it, their similarity drops to the unrelated-pair baseline. This signature has been verified directly in the HDC framework’s numerical studies (synthetic relation encodings, and time-reversal of real speech token sequences, where the role-free control produced cosine +1.000 exactly); the appendix reuses it as a wiring self-test.

3.2 What $\hat{K}$ and $\hat{E}$ become

With frame codes fixed, $\hat{K}$ is a readout from transformer latents to $\mathbb{H}_D$: a linear or shallow map from selected residual-stream or head-output subspaces to a frame hypervector, trained by the losses of Section 4. $\hat{E}$ maps frame codes to linguistic-side codes; in the simplest useful instantiation, linguistic objects are also hypervectors (bags of typed morphosyntactic features, construction types, and linearization cues, bundled with their own roles), and $\hat{E}$ is again a small map. The composite $\hat{\Lambda}$ is then measurable end to end. Nothing prevents richer instantiations— $\hat{E}$ into a symbolic MeTTa/Atomspace layer, with hypervectors as the associative regime between—but the all-hypervector version is the one that makes every CGCCT loss a vector computation, and it is the version the small-scale programme tests first.

3.3 Probes as algebra, and calibration as capacity

A feature probe in this object language is not a trained classifier but an algebraic readout: the presence of feature a in a frame or linguistic code F is scored by $\langle F, r_{\text{feat}} \otimes h_a \rangle / D$ against a margin. This has two consequences.

First, the closure losses become linear inequalities on similarities: an implication $a \Rightarrow b$ is the penalty $[\langle F, p_a \rangle - \langle F, p_b \rangle]_+$ on codes asserted to satisfy a , with $p_x = r_{\text{feat}} \otimes h_x$. No probe training, no probe drift, no probe overfitting to the network being audited.

Second—and this is the principal guarantee transfer—the δ -calibration that CGCCT’s closure-preservation argument assumes is now a theorem schema rather than an assumption. Probe reliability is a cleanup-margin condition, and Eq. (2) prescribes the dimension at which it holds: for feature inventory size M , effective bundle load k (how many terms are superposed in the codes being probed), and feature-dictionary coherence ν , calibration to tolerance ε needs $D \gtrsim k(1 - \nu)^{-1} \log(M/\varepsilon)$ up to constants. The appendix’s first stage measures exactly this curve on codes with planted ground truth.

Conjecture 1 (Hierarchies live in the high-coherence regime). *Implicational hierarchies are, by construction, chains of features whose realizations differ minimally: adjacent hierarchy positions share most of their conditioning structure. When such features are encoded as role-bound codes over shared constituents, adjacent positions are near-duplicate dictionary entries in the sense of the HDC*

coherence analysis, with $1 - \nu = \Theta(1/k)$. Consequently the dimension needed to probe hierarchy-adjacent features reliably scales as $k^{3/2} \log M$ (sign codes) to $k^2 \log M$ (linear/phaser codes), not $k \log M$, and closure-loss gradients computed below that dimension are dominated by probe confusion rather than by genuine violations.

If the conjecture holds, it has a practical edge: it predicts a specific, testable failure mode of naively implemented closure training (spurious closure “violations” that vanish as D grows on the predicted schedule), and it says that hierarchy probes should be the first place to spend dimension or separation (summary channels, margin losses) rather than the last. Stage P1 of the appendix tests it directly.

4 Naturality as computable algebra

4.1 Implementing V_c

The naturality loss compares $\widehat{\Lambda}(U_c z)$ with $V_c \widehat{\Lambda}(z)$. Its practical obstacle has been the right-hand side: V_c must be an executable transform on linguistic objects, and d must be a meaningful metric. With hypervector frames and linguistic codes, a useful family of V_c is available immediately:

- **Identity** ($V_c = \text{id}$): paraphrase, register change, and translation pairs assert that the *frame* code is preserved while the linguistic code varies (paraphrase) or that designated linguistic features vary while frame content is fixed (translation). These are contrastive constraints requiring no annotation beyond the pairing.
- **Known rebind**: argument alternations—active/passive, dative shift, causative/inchoative, symmetric-predicate exchange—correspond to *specific* role remappings. The transformed sentence’s frame code should equal the original’s with an algebraic slot swap applied; its linguistic code should differ by known feature substitutions (voice, case-frame). The right-hand side of the naturality square is computed, not learned.
- **Closure addition**: contexts that add material (a temporal adjunct, a further conjunct) correspond to adding bound terms; naturality asserts the old content survives unbinding after the addition, which is precisely the HDC crosstalk discipline with the addition counted in k .

The HDC framework’s role-consistency loss—for content c appearing under roles r_a and r_b , penalize $\|r_a^\dagger \otimes z_a - r_b^\dagger \otimes z_b\|$ —is exactly the naturality square for the known-rebind V_c with a one-slot remap. CGCCT names the functorial family; hypervectors make each member differentiable and cheap.

4.2 Naturality defect, decomposed

Because both sides of the square are vectors, the naturality defect $d(\widehat{\Lambda}(U_c z), V_c \widehat{\Lambda}(z))$ decomposes along the pipeline: a *frame-level* defect $d(\widehat{K}(U_c z), v_c \widehat{K}(z))$ (with v_c the frame-side transform) and an *externalization* defect contributed by $\widehat{E}$. This mirrors the factorization-defect discipline of the HDC framework—and inherits its warning. A small naturality defect on an alternation battery certifies the batteries’ query family, nothing more; it is not a certificate for open-ended generation, and the appendix’s report format requires the query family to be stated next to every defect number.

4.3 Where supervision comes from

The economically important observation is that the supervision for the frame layer largely already exists in unannotated or lightly annotated form: paraphrase corpora and backtranslation give identity squares; templated alternation batteries (generated programmatically over verb frames) give known-rebind squares with exact right-hand sides; morphological analyzers give linguistic-side feature codes for closure probes in richly inflected languages. None of this requires full semantic parsing, which is what makes a GPT-2-small-scale programme feasible at all.

5 Footprints, mediators, and double-counting in hypervector space

5.1 Footprint codes

CGCCT audits modules by their do-footprints: for module m , the tuple of its causal influences on the output, on frame content, on linguistic features, and on context routing. Bundle these into a footprint hypervector

$$Z_m = r_y \otimes z_{y,m} \oplus r_F \otimes z_{F,m} \oplus r_L \otimes z_{L,m} \oplus r_{\text{ctx}} \otimes z_{\text{ctx},m} \oplus r_{\text{sig}} \otimes H_m, \quad (5)$$

where each $z_{\cdot,m}$ encodes the corresponding influence profile (e.g. a bundle of feature atoms weighted or thresholded by patched-ablation effect sizes) and H_m is the module’s content signature. Then:

- **Double-counting is coherence.** Two modules with redundant causal roles have high $\langle Z_m, Z_n \rangle / D$; the double-counting penalty is a coherence penalty on the footprint dictionary $\{Z_m\}$, i.e. exactly the quantity ν that the HDC framework already measures and regularizes, computed over module space instead of state space. The split/merge recipe (merge modules with redundant footprints, split modules whose footprint is a mixture) becomes cleanup clustering over $\{Z_m\}$, and it scales: with hundreds of modules, redundancy detection is an approximate-nearest-neighbor query, not a quadratic audit.
- **Mediator structure is the bridge budget.** The CGCCT mediator rule—no dense $A \leftrightarrow B$ coupling; route through a sparse M —is the HDC bridge-microcolumn rule: cross-module content must pass through typed directed relation terms $r_{\text{rel}} \otimes r_{\text{src}} \otimes r_{\text{tgt}} \otimes c$, whose count k_B is measured and budgeted like any local load. The residual quantale value $M^* = A \Rightarrow B$ (the weakest mediator sufficing for the observed dependency) acquires a concrete estimator: the smallest typed relation load whose removal severs the measured do-dependency. Mediator *leakage* likewise becomes addressable: content flowing through the mediator channel that fails to decode as typed relations is, by construction, leakage, and it is enumerable by cleanup rather than inferred by probing.

5.2 Commutator weights and the confusion graph

The commutator loss weights context pairs by frame and linguistic overlap $\omega_{cd} = \text{Overlap}_F(c, d) \cdot \text{Overlap}_L(c, d)$. Encode each context by a support code $H_{S_c} = \text{Normalize}(\bigoplus_{m \in S_c} r_m \otimes H_m)$ together with its frame and linguistic codes; the overlaps are similarities, and the sparse confusion graph over contexts—which pairs are worth regularizing at all—is an approximate-nearest-neighbor graph over these codes, built incrementally. The HDC framework’s episodic support memory (task, support, certificates, outcome, audit) is precisely the store this graph wants to be built from, which means the continual-learning side of CGCCT can reuse ColBaC-style memory infrastructure unchanged.

6 The weakness quantale gets a cost coordinate

CGCCT’s selection principle is multi-objective: among structures matching the data, prefer the weakest, where weakness is valued in a product quantale over fit, retention, commutator defect, leakage, double-counting, and cost. The HDC capacity proxy $\mathcal{C}_{\text{HDC}}$ of Eq. (3) is a natural occupant of the cost-and-confusion coordinates: it is a single measurable scalar that increases with effective load, confusable alternatives, and dictionary coherence, and decreases with margin—exactly the resource profile that “prefer one sparse mediator over redundant correlation” implicitly prices. Read this way, the HDC-guided training objective of the capacity framework is a scalarization of the CGCCT weakness order restricted to its representational coordinates, and the two selection procedures can share bookkeeping: the same measured $(k^{\text{eff}}, M^{\text{conf}}, \nu, \Delta)$ tuple feeds both.

Two practical corollaries. First, weakness comparisons between candidate module structures become cheap enough to run inside a training loop, because every coordinate is a vector computation over codes already being maintained. Second, the quantale’s monotonicity requirements (composition cannot improve hom-values) are respected by construction, since similarity under composition of algebraic operations can only degrade—a small point, but one that keeps the enriched-category story honest rather than decorative.

7 What is measurable at small scale

The merged framework’s quantities separate cleanly by the cheapest substrate adequate to measure them. The ladder below is the design basis of the appendix; the general principle is that *instrument validation precedes hypothesis testing*: before any comparison of trained models, the probes, naturality squares, and footprint machinery must be shown to behave as the capacity theory predicts on systems where ground truth is available.

Quantity	Cheapest substrate	How
Probe calibration δ vs D ; closure-violation floor; coherence regime of hierarchy probes (Conjecture 1)	synthetic-grammar transformer (6 layers)	plant implicational hierarchies by construction; measure calibration and spurious-violation curves against Eq. (2)
Naturality defect of a frozen model; alternation squares	GPT-2 small (frozen)	templated alternation batteries; $\hat{K}$ trained as a readout only; known-rebind right-hand sides
Do-footprints; double-counting as coherence; mediator budgets; known-answer circuit rediscovery	GPT-2 small (frozen)	patched ablations over 144 heads; footprint codes; the indirect-object-identification circuit as a planted answer
Commutator defect; forgetting; the CGCCT four-model comparison in miniature	GPT-2 small + adapter context streams	sequential context streams with reversed-order controls
Stability coefficients σ_u over training	Pythia-160M checkpoint series	universal-satisfaction probes evaluated across public checkpoints; GPT-2 cannot supply this
Cross-linguistic closure; morphological hierarchies	small multilingual model	English is morphologically poor; person/case hierarchies are barely observable in it regardless of method

Three things cannot be measured small, and the programme does not pretend otherwise: whether category-guided training preserves language-model quality at scale (small models are undertrained, so interference patterns do not transfer); whether the frame bottleneck’s defect stays low for open-ended *generation* (a weak model generates badly for unrelated reasons, confounding the defect measurement); and genuine typological breadth (a data problem, not a compute problem). The sharp prediction that *is* fully testable small is CGCCT’s own: matched validation loss with lower commutator defect, double-counting, and forgetting for the guided models. A clean positive there is the evidence that justifies scaling; a clean negative, with validated instruments, is a real result about the approach.

8 Discussion

What the merger buys. Concretely: an object language for Frame_Q whose morphisms execute; naturality squares with computed right-hand sides and free supervision; closure probes whose calibration is a capacity theorem with a dimension prescription; footprint auditing that scales as similarity search; a shared cost coordinate for weakness selection; and reuse of episodic support memory for the confusion graph. In the other direction: a principled inventory of which HDC probes and invariances a language system should be trained toward, replacing an ad hoc loss menu with one indexed by universals and alternations.

What it does not buy. The existence of a low-defect frame factorization of transformer language states is assumed by both frameworks and guaranteed by neither. The merger converts this assumption into measurable quantities with predicted parameter dependence, which is the most that honest methodology can do before the measurements are made. The small-scale programme is therefore front-loaded with instrument validation and known-answer checks, and its stop conditions treat a well-measured null as a deliverable rather than a failure.

Relation to the broader programme. The synthesis slots into an existing lineage: the causal-coding and HBCML/ColBaC line supplies the modular substrate and teacher machinery; the HDC capacity line supplies the representational algebra and its budget; CGCCT supplies the linguistic supervision grammar. The appendix’s programme is deliberately written in the same staged, gated, agent-executable format as the causal-fibres and predictive-coding programme documents, so that the same operational tooling—and the same standards for what counts as a validated instrument—carry over.

A Agent testing programme at GPT-2-small scale

This appendix is written for autonomous coding agents. Run stages in order. Each stage has a purpose, procedure, gates, budget, and stop conditions. **A cleanly measured null is a successful outcome; the only failure is a corrupted positive.** Fix only what blocks execution; record every deviation. Frozen design choices (Section A.10) may never be changed.

A.1 Mission and non-goals

The programme validates the measurement instruments of the merged CGCCT–HDC framework and then runs the smallest honest version of the CGCCT four-model comparison. Non-goals: demonstrating at-scale quality preservation; open-ended generation quality; typological breadth

beyond English plus one templated multilingual battery. Do not benchmark generation samples; do not report perplexity improvements as the headline (the framework predicts *parity*, not improvement, on validation loss).

A.2 Environment and substrates

- One GPU ≥ 16 GB. Python ≥ 3.10 ; `torch`, `transformers`, `datasets`, `numpy`, `einops`. Pin versions in the report.
- Substrates: (i) a 6-layer synthetic-grammar transformer trained from scratch (reuse the causal-fibres E-series harness where available); (ii) `gpt2` (124M) frozen, plus LoRA adapters where training is called for; (iii) optionally `EleutherAI/pythia-160m` checkpoint series for stage P6; (iv) tiny-shakespeare only as a plumbing smoke test, never as an evidential substrate.
- All hypervector machinery in float32 bipolar MAP with a *fixed* per-dimension tie-breaking convention (deterministic encoder). Randomized tie-breaking is a known artifact source; do not use it.

A.3 Stage P0: infrastructure and self-tests

Purpose. Stand up hooks, the HDC probe library, and wiring self-tests before any measurement.

Procedure.

1. Implement the frame-code library: Eq. (4) encode, role rebind, slot swap, role-mask restrict, algebraic probes, cleanup with margins, batched resonator. Property-test each against hand-computed cases.
2. Orientation self-test: encode argument-exchanged two-role frames with and without the target-argument permutation. Gate: without, cosine = 1.000 exactly; with, cosine within ± 0.05 of the unrelated-pair baseline measured on random frame pairs.
3. Probe self-test on planted bundles: k role-bound features, probe each; reproduce the single-step capacity curve (accuracy vs D at fixed k, M) qualitatively.
4. Transformer hooks: residual-stream and per-head output capture on `gpt2` at every layer; verify shapes and determinism (two runs, same seed, identical captures).
5. Tiny-shakespeare smoke: end-to-end run of the P2 pipeline on a char-level model, checking only that nothing crashes and artifacts are written. Its numbers are not evidence; do not report them as such.

Budget. < 1 GPU-hour. **Stop** if any self-test gate fails after two fix attempts.

A.4 Stage P1: capacity-law validation on synthetic grammar (instrument validation)

Purpose. Verify that probe calibration follows Eq. (2) with ground truth available, and test Conjecture 1.

Procedure.

1. Generate a synthetic language from a PCFG with a planted implicational hierarchy of $H \in \{4, 6\}$ features (feature j present only if $j' < j$ present), realized as morpheme-like markers; train the 6-layer transformer to convergence on next-token prediction.
2. Train $\hat{K}$ as a linear readout from the residual stream to frame codes over the planted roles and features (supervision available by construction).
3. Measure probe calibration $\delta(D)$ for $D \in \{256, \dots, 16384\}$: probe each feature on held-out states; calibration is the gap between probe scores on true positives and true negatives, and the spurious-closure-violation rate is the fraction of held-out states flagged as violating a planted implication.
4. Construct hierarchy-adjacent versus hierarchy-distant probe pairs and measure their dictionary coherence ν ; fit the required- D scaling for adjacent pairs.

Gates.

1. Calibration improves monotonically in D and the required D at fixed tolerance scales linearly in measured $k/(1 - \nu)$ within a factor of 2.
2. Spurious violation rate falls below 1% at the D predicted by the fitted law, and adjacent-pair required D exceeds distant-pair required D by a factor consistent with the $k^{3/2}$ – k^2 regime (report the fitted exponent; the conjecture is falsified if it is ≈ 1).

Budget. 2–6 GPU-hours. **Stop** the whole programme if Gate 1 fails: closure training downstream would be uninterpretable.

A.5 Stage P2: frozen-model naturality on GPT-2 small

Purpose. Measure the naturality defect of a *pretrained, frozen* model under known-rebind squares—the cheapest probe of whether frame structure is present to be found.

Procedure.

1. Build templated alternation batteries ($\geq 2,000$ items each): active/passive, dative shift, and a symmetric-predicate exchange control (where the frame genuinely changes); plus a paraphrase battery from backtranslation or a public paraphrase set.
2. Train $\hat{K}$ (linear, then 2-layer MLP as a robustness check) from frozen **gpt2** states to frame codes, using ONLY contrastive supervision: paraphrase pairs pull frame codes together; alternation pairs are tied by the algebraic slot swap; random pairs push apart. No parsed frames are given.
3. Measure: (a) frame-level naturality defect $d(\hat{K}(z_{\text{alt}}), \text{swap}(\hat{K}(z)))$ per battery; (b) role decodability: unbind agent/patient roles from $\hat{K}(z)$ and cleanup against a filler dictionary built from the battery’s noun inventory; (c) the symmetric-predicate control, where the swap should NOT hold.
4. Controls: shuffled-pairing control (defect should rise to the random level); layer sweep (report defect vs layer); untrained-transformer control (random **gpt2** initialization; role decodability should collapse toward chance).

Gates.

1. Role decodability on held-out items beats chance by $\geq 5\times$ and beats the untrained-model control by $\geq 2\times$.
2. Alternation defect is significantly below the shuffled control, and the symmetric-predicate control shows the predicted *absence* of invariance.

Budget. 4–10 GPU-hours. If gates fail with the linear $\hat{K}$ but pass with the MLP, report both; if both fail, this is a substantive negative about extractable frame structure in **gpt2**—report it and consult the operator before proceeding to P5 (P3 remains worthwhile regardless).

A.6 Stage P3: footprints, double-counting, and the IOI known-answer check

Purpose. Validate the footprint machinery against a circuit whose answer is known.

Procedure.

1. Implement patched-ablation do-footprints for all 144 attention heads of **gpt2** on (i) the indirect-object-identification (IOI) task distribution and (ii) a neutral corpus slice; encode footprints as Eq. (5) codes.
2. Cluster footprint codes by cleanup/ANN; compute the double-counting coherence matrix over heads; extract the top redundancy groups.
3. Known-answer check: the head groups identified for the IOI distribution must substantially recover the published IOI circuit structure (name-mover heads, S-inhibition heads, duplicate-token heads), including at least one redundancy group corresponding to the known backup name-mover phenomenon. *Verify the reference decomposition against the literature before scoring—the drafting model’s recall of the circuit membership must not be treated as ground truth.*
4. Mediator probe: estimate minimal typed-relation budgets k_B between the recovered groups and compare against direct dense-coupling estimates.

Gates. Head-group recovery overlaps the verified published circuit with Jaccard ≥ 0.4 at the group level; the redundancy (double-counting) detector flags the known backup heads. If recovery fails, debug the footprint estimator on synthetic planted circuits (two heads with forced redundant roles) before concluding anything.

Budget. 6–12 GPU-hours (ablation sweeps dominate).

A.7 Stage P4: closure probes on natural text

Purpose. Move closure measurement from planted to natural hierarchies, within English’s limits.

Procedure. Using $\hat{K}$ from P2 and morphosyntactic feature codes from an analyzer or tagger: measure closure-violation rates for agreement-related implications and animacy/definiteness effects on held-out corpus text, at the D prescribed by the P1-fitted law; include a deliberately false implication as a negative control (its “violation rate” must be high). Optionally repeat one battery on a small multilingual model with a richly inflected language to observe a hierarchy English cannot show. **Gate.** True implications violate at $< 5\times$ the P1-predicted probe-noise floor; the false-implication control violates at $> 20\times$ that floor. **Budget.** 2–4 GPU-hours.

A.8 Stage P5: the four-model comparison in miniature

Purpose. The smallest honest version of CGCCT’s central experiment: does category-guided training deliver matched validation loss with lower commutator defect, double-counting, and forgetting?

Procedure.

1. Define $C = 6$ context streams over **gpt2**-scale data (e.g. genre/domain slices), trained sequentially via LoRA adapters plus a small shared trunk unfrozen; two stream orders (forward, reversed).
2. Train four variants with matched budgets: (1) baseline; (2) modular routing without linguistic losses; (3) causal-coding losses (mediator, double-counting, commutator with HDC overlap weights) without frame supervision; (4) full: (3) plus frame-layer naturality and closure losses from P2/P4 machinery.
3. Metrics per variant: validation loss per stream; commutator defect on the P3-derived confusion graph; double-counting coherence; forgetting (loss regression on earlier streams); naturality defect; closure-violation rate; and the weakness-quantale coordinates including $\mathcal{C}_{\text{HDC}}$.
4. Three seeds minimum; report per-seed numbers, not just means.

Gates and reading. The framework’s prediction is: validation loss within noise of baseline for (3) and (4), with monotone improvement (1)→(4) on commutator defect, double-counting, and forgetting, and with (4) beating (3) on naturality and closure specifically. Any variant that improves auxiliary metrics by *sacrificing* validation loss beyond noise has failed the parity requirement and must be reported as such. **Budget.** 1–3 GPU-days; this is the expensive stage and is gated on P1–P2 passing.

A.9 Stage P6 (optional): stability coefficients on Pythia

Evaluate the P2/P4 probe batteries across the public **pythia-160m** checkpoint series; report, per universal, the training step at which probe satisfaction stabilizes (the empirical σ_u), and whether early-stabilizing universals correspond to lower-variance module footprints in P3’s sense. Budget: 3–6 GPU-hours (inference only). This stage is descriptive; it has no pass/fail gate.

A.10 Frozen choices and allowed knobs

Never change: the deterministic tie-breaking convention; the identity of the four P5 variants and their matched budgets; the known-answer references of P0 and P3 (verified against literature, then frozen); scoring definitions (calibration, defect, violation rate) once P1 completes; the parity requirement in P5.

May change with a one-line report note: D grids; battery sizes; LoRA rank; learning rates (halve on divergence, clip gradients at 1.0); layer choice for $\hat{K}$ (sweep and report); number of streams C (4–8).

A.11 Report format

Produce **RESULTS.md** containing: environment and pins; per-stage gate outcomes with the measured numbers beside their thresholds; every deviation in one line each; the P1 fitted capacity-law exponents; the P2 defect table with all controls; the P3 circuit-recovery Jaccard and redundancy

findings; the P5 four-by-metrics table per seed; and a closing paragraph answering, in order: did the instruments validate; does frozen GPT-2 contain extractable frame structure; did guided training achieve parity-plus-hygiene; and what single follow-up is most informative. Attach all JSON artifacts.

A.12 Global stop conditions

Stop and report (partial results are a deliverable) if: P1 Gate 1 fails; P2 both-readout failure is confirmed and the operator directs a halt; cumulative compute exceeds 5 GPU-days; or any fix would touch a frozen choice.

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